

1. THE PROBLEM, AND WHAT MAKES IT HARD

We predict `energy_usage` for a campus building from its identity, time, weather, occupancy and its preceding reading. The headline difficulty is not finding signal — `previous_usage` alone correlates 0.945 with the target, and simply copying it forward scores RMSE 6.40. The difficulty is that *the test set is systematically harder than the training set*, in a way that silently corrupts how a team measures its own progress.

2. TWO FINDINGS THAT SHAPED EVERYTHING

(a) This is not a time series, despite looking like one. With `hour`, `month`, a lag feature and a tidy ID split, a time-aware split is the natural response. We tested rather than inherited that: there is no timestamp, so ordering is undefined; every ID-feature correlation falls within ± 0.016 ; and an ID-ordered holdout scored 3.606 against 3.560 for random 5-fold — inside the ± 0.071 fold noise. We use random K-fold, keeping 20% more training data than a time split would allow.

(b) Standard validation is 11.2% optimistic here.

Every gap-prone feature is missing 4–5× more often in test than in train. 18.2% of test rows carry at least one gap versus 5.8% of training rows; for rows with two or more gaps the split is 7.7% versus 0.14% — a 55× difference. A model validated on training-like data is therefore graded on an easier problem than it will face. We measured the bias directly: one fitted model, same validation rows, scored 3.560 as-is and 3.960 once the fold was degraded to test-level missingness.

The consequence is worse than an inflated number. Models degrade at different rates, so naive validation can *misrank* them. On our own earlier candidates, a Ridge model appeared 0.20 RMSE better than gradient boosting and was actually 0.08 worse under test conditions. All model selection in this work is therefore made on RMSE with test-level missingness injected into every validation fold, fixed before any model was tuned.

3. APPROACH

Features. Exploration showed buildings differ in *shape*, not merely in level: mean usage spans 30.8 (Administration) to 79.2 (Science), while variability ranges from sd 6.5 (Residential, highly regular) to 19.1 (Lecture Halls, volatile). We encode this explicitly — per-building intercept, 24-hour profile, weekend offset, and per-building slopes on occupancy, temperature and previous usage (366 features) — plus cyclical hour encoding and per-column missingness indicators.

Imputation. This proved the single most valuable decision. A median is not merely imprecise: the building slopes are products `is_building × value`, so one missing input collapses the entire building-specific relationship for that row. We instead train a dedicated LightGBM per gap-prone column, predicting it from building, hour and the remaining drivers, then regenerate the interaction terms. Crucially the imputer predicts *features from features* — the target is never involved — so it can legitimately be fitted on the training and test features together, giving it 11,000 rows instead of 8,000 and direct exposure to the distribution it must impute into. Tested on identical folds:

IMPUTATION STRATEGY	RMSE
Global median (standard practice)	3.852
Building-level median	3.614
MICE / iterative imputer	3.562
Sub-model (LightGBM per column)	3.339
Sub-model, fitted transductively	3.209

Augmentation. Training contains roughly 11 multi-gap rows against ~231 in test, so the model has almost no exposure to the regime it is graded on. Degrading the training data to test-level missingness improved every model tested (CatBoost by 0.265) and halved fold variance. This uses only the observed missingness *rate* of the test features — no test labels are involved.

4. MODEL SELECTION, AND REJECTING THE ENSEMBLE

Ridge with explicit building interactions outperformed every gradient-boosted alternative. The structure here is close to additive — a building's base level, its own daily profile, and near-linear responses to its drivers — which a

regularised linear model fits directly and extrapolates from sensibly, while trees must approximate smooth relationships with axis-aligned splits.

MODEL (REALISTIC VALIDATION)	RMSE	MAE
Ridge + building interactions	3.199	2.507
CatBoost	3.487	2.708
LightGBM	3.548	2.759
XGBoost	3.582	2.783
4-model weighted stack (nested)	3.261	—

We built the ensemble and rejected it. Scored conventionally it appeared best (3.226), but those weights were fitted on the predictions being scored. Nested — weights learned on inner folds, evaluated on an untouched outer fold — it scored 3.261, worse than the single Ridge. Its advantage was 0.035 RMSE of self-deception: the same error we had diagnosed in the validation protocol, caught in our own work.

5. RESULT

3.199 RMSE	2.507 MAE	0.970 R²
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Against references of 18.45 (predict the mean), 13.22 (building mean) and 6.40 (copy `previous_usage`, no model). All figures are under injected test-level missingness; conventional validation reports materially better numbers we do not consider trustworthy.

6. ERROR ANALYSIS

Residuals are centred and symmetric, with no bias across the prediction range. Error tracks each building's intrinsic volatility: Lecture Halls hardest, Residential easiest. Much is irreducible, but it means a campus-wide average understates uncertainty on exactly the buildings an operator would most want warning about. Error also rises with missing-input count, far more gently than median imputation produces.

7. COVARIATE SHIFT: DETECTED, AND DELIBERATELY NOT CORRECTED

A classifier trained to separate training rows from test rows reaches AUC 0.688 — the two sets are distinguishable. This is not an artefact of the missingness: removing the missingness indicators leaves the AUC unchanged (0.689), and restricting to complete cases, where no gaps exist at all, still gives 0.654. The shift is in the feature values themselves (test rows are busier and carry higher previous usage).

We applied the textbook correction — importance weighting by the train/test propensity ratio — in four variants. Every one made the model worse (3.28–3.51 vs 3.23 unweighted). Covariate shift only induces bias when the model is misspecified: if $P(\text{energy} \mid \text{features})$ is stable, reweighting removes no bias and adds variance, exactly the pattern observed. The physics of a building does not change because the sample contains busier ones. We report the shift as a measured property and deliberately leave it uncorrected.

8. LIMITATIONS AND NEXT STEPS

- No timestamps, so genuine lag or rolling features are impossible; `previous_usage` is the only history available.
- Imputation error is invisible downstream. Missingness flags let the model discount estimated values; propagating imputation uncertainty would be the principled fix.
- Augmentation rates are calibrated to the public test set. If the private set is harsher they are slightly off, though this should still degrade more gracefully than median-based alternatives.
- High-variance buildings dominate residual error. Per-building or mixed-effects models are the next step, alongside a cost-weighted error view — a 5-unit error on a 30-unit admin block matters more proportionally than on an 80-unit science building.
- Thirty-three variations were tested; four were kept. Rejected on evidence: log and delta target framings, extra interaction families, splines, residual boosting, target encoding, per-building models, specialist routing for gap rows, imputer tuning, pseudo-labelling and importance weighting. We read this as a sign the remaining error is largely irreducible — with `previous_usage` at $r = 0.945$ there is limited signal left to extract.